# Acceleration means velocity is changing

GUIDED NOTES / use with the projected diagram Name _____

## A SAY THE WORD

### Acceleration / ES: aceleración / PT: aceleração

The rate velocity changes: how much velocity is gained or lost each second.

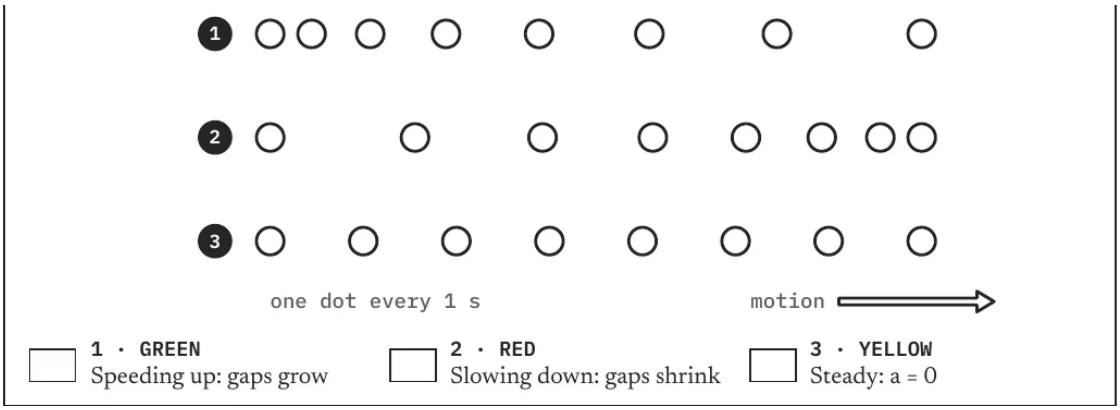
Also on your note page: Speeding up, Slowing down, Turning.

## B READ THE RELATIONSHIP

### Acceleration tells how quickly velocity changes.

Average acceleration = change in velocity ÷ time

## C TRACE THE ORIGINAL FIGURE




Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

### A cart's velocity changes from 0 to 8 m/s in 4 s. Find its acceleration.

**Given** $v_i = 0$ m/s $v_f = 8$ m/s $\Delta t = 4$ s

**Rule** $a = \Delta v \div \Delta t$

**Work** $\Delta v = 8 - 0 = 8$ m/s $a = 8 \div 4$

**Answer** 2 m/s² gains 2 m/s each second

## E SAVE FOR THE FINAL MEETING

If XJ's velocity readings change from one night to the next, it is accelerating, and a forecast that assumes steady velocity will miss.

# Read the physics: Acceleration

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

Compare velocity before and after. A change in speed or a turn is a change in velocity. Divide the signed change in velocity by the time it took.

## 02 HOW IT WORKS

Read m/s² as a change in meters per second each second. Look at dot spacing at equal times: growing gaps mean faster motion; shrinking gaps mean slower motion.

## 03 WHAT TO CHECK

Compare velocity and acceleration. In a straight line, matching directions mean speed grows. Opposite directions mean speed falls until the object stops or reverses.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

A cart's velocity changes from 0 to 8 m/s in 4 s. Find its acceleration. 2 m/s² gains 2 m/s each second.

## 05 CONNECT TO XJ

If XJ's velocity readings change from one night to the next, it is accelerating, and a forecast that assumes steady velocity will miss.

SAY IT BEFORE YOU WRITE

Can acceleration be negative while speed increases? The velocity changes from ___ to ___. Each second it changes by ___.

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Acceleration

ES: aceleración / PT: aceleração

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain acceleration to a friend?<br>In my words, this means ___. | A cart follows a bend at steady speed. Why is it accelerating?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Positive is right. A cart goes from −2 to −6 m/s in 2 s. What acceleration fits?<br>My example shows ___. | A cart travels straight at −6 m/s the whole time. Does the minus sign mean it is accelerating?<br>I would change ___ because ___. |

## Speeding up

ES: aumento de rapidez / PT: aumento de rapidez

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain speeding up to a friend?<br>In my words, this means ___. | A cart changes from −2 to −5 m/s. What happens to its speed?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Draw three equal-time dots for a cart speeding up to the right.<br>My example shows ___. | A cart moves right with acceleration to the left. Is it speeding up before it turns?<br>I would change ___ because ___. |

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Slowing down

ES: disminución de rapidez / PT: redução de rapidez

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain slowing down to a friend?<br>In my words, this means ___. | Velocity changes from −5 to −2 m/s. What happens to the speed?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Draw velocity and acceleration arrows for a bike braking while moving right.<br>My example shows ___. | "Negative acceleration always means slowing down." What happens if velocity is negative too?<br>I would change ___ because ___. |

## Turning

ES: cambio de dirección / PT: mudança de direção

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain turning to a friend?<br>In my words, this means ___. | A rider keeps the speedometer at 4 m/s around a bend. Which part of velocity changes?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How would you draw two velocity arrows as a cart turns from east to north at steady speed?<br>My example shows ___. | A cart brakes on a straight line and still moves east. Has it turned?<br>I would change ___ because ___. |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Draw the before and after velocity arrows (m/s). Mark the change in velocity (m/s). Divide by time to label the acceleration (m/s²).

Labels you can use: before · after · change.

Before

After

**Tell a partner**

The velocity changes from ___ to ___. Each second it changes by ___.

# Choose the speed alert

07

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

**Our goal:** Use a velocity change to find acceleration.

**CHOOSE ONE JOB TOGETHER**

**TELEMETRY ENGINEER** ☐

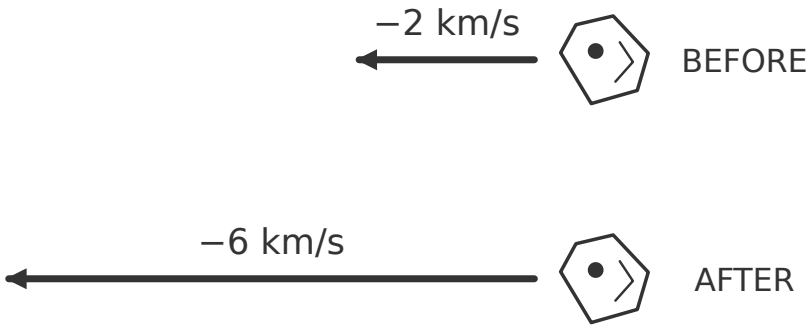
Radio the average acceleration with its sign and unit. Label both velocity arrows so the crew can check it.

**REPLAY COMMENTATOR** ☐

Narrate the two-second change. Point to the arrow lengths to explain whether XJ sped up or slowed down.

**Word help:** velocity / acceleration / speeding up

Two signed XJ velocities are **−2** and **−6 km/s**, two seconds apart. The control room must choose its speed alert.




**01 OUR FIRST CALL**

**Which speed alert should the control room use? Support it with the arrows.**

The speed ___ because the arrow ___.

**02 OUR REVISED CALL**

**Wait for the reveal. Keep your first call.**

New clue: _____

Check the proposed banner. Give a corrected report using the two velocities.

Now we would say ___ because ___.

**Switch speakers.** Which arrow shows the larger speed?

| ☐ GREEN | ☐ RED | ☐ YELLOW | ☐ BLUE | ☐ PURPLE | ☐ ORANGE | ☐ PENCIL |
|---|---|---|---|---|---|---|
| speeding up | slowing down | steady / balanced | position / displacement | velocity | forces | other marks |